PsrPopPy Documentation

Release 1

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ONE

INTRODUCTION

PsrPopPySuper is a modernized Python3 port of Sam Bates' PsrPopPy code (https://github.com/samb8s/psrpoppy). This project retains the original code's functionality while updating interfaces and adding newer electron models (for example NE-2025) alongside NE-2001 and LMT85.

Contributions are welcome via the project's GitHub repository.

CHAPTER

TWO

INSTALLATION

To get started with PsrPopPySuper there are a few steps you'll need to go through.

PsrPopPySuper is currently supported on Linux and Mac OS X, and for full feature support, it is recommended to install the Matplotlib package and either use Python versions >3.11, or install the argparse module.

2.1 Download the package

The source for PsrPopPy can be downloaded from GitHub from the PsrPopPy page. The source will contain the Python modules and scripts needed both for basic and advanced use.

2.2 Installing the package

PsrPopPy now supports standard Python package installation using pip. This automatically handles the Fortran compilation and installation.

2.2.1 Prerequisites

- Python 3.11 or later
- gfortran compiler (for Fortran library compilation)

On macOS, install gfortran using Homebrew:

```
brew install gcc
```

On Linux, install gfortran using your package manager:

```
# Ubuntu/Debian
sudo apt-get install gfortran

# CentOS/RHEL/Fedora
sudo yum install gcc-gfortran  # or dnf install gcc-gfortran
```

2.2.2 Installation

From the source directory, run:

```
pip install .
```

This will: 1. Compile the Fortran libraries using gfortran 2. Install the Python package 3. Install command-line scripts (dosurvey, populate, evolve)

For development or user installation:

```
pip install -e .  # editable install
pip install --user .  # user install
```

2.3 Legacy Fortran Compilation (if needed)

Although PsrPopPySuper is a Python-based package, some of the algorithms have been kept in their native FORTRAN for speed and ease of programming (e.g. the NE2001 electron model, coordinate conversion...). The pip installation now handles this automatically, but if you need to compile manually:

From the base directory:

```
cd psrpoppy/fortran
```

To use `make`, edit `makefile.<OSTYPE>` and ensure that the gf variable points to the location of your gfortran compiler. Then simply type `make`. All being well, four .so files will be generated.

Failing this, edit either `make_mac.sh` or `make_linux.sh`, depending upon your system, so that the `gf` variable points to your local gfortran/f77 compiler. Running the script:

```
bash make_<os>.sh
```

CHAPTER

THREE

COMMAND-LINE SCRIPTS

3.1 populate.py

-n <number of pulsars>
: Required: Number of pulsars to generate; or to detect in a survey

-o <output>
: Output file name for population model (def=populate.model)

-asc <ascii output file name>
: Output file name for ascii output file (def=populate.txt)

-surveys <SURVEY NAME(S)>
: List of surveys to use when trying to detect pulsars (default=None)

-z <scale height>
: Scale height of pulsars about Galactic plane, in kpc (default=0.33)

-w <width>
: Pulse width to use when generating pulsars (default=0, use beaming model)

-si <SImean SIsigma>
: Spectral index mean and standard deviation (default=-1.6, 0.35)

-sc <scatter index>
: Spectral index of scattering law to use (default=-3.86, Bhat et al model)

-pdist <distribution name>
: Distribution type for pulse periods (default=lnorm)

  Supported: 'lnorm', 'norm', 'cc97'

-p <mean stddev>
: Mean and standard deviation to use in period dist 'lnorm' or 'norm' (def=-2.7, 0.34)

-rdist <radial model>
: Model to use for Galactic radial distribution of pulsars

  Supported: 'lfl06', 'yk04', 'isotropic', 'slab', 'disk'

-dm <Electron model>
: Model to describe the Galactic electron distribution

  Supported: 'ne2025', 'lm98', 'ne2001'

-gps <fraction 'a'>

Add <fraction> pulsars with GHz-frequency turnovers with index a

-doublespec <fraction alpha1>

Add <fraction> pulsars with low-frequency (below 1GHz) spectral index of alpha1

--nostdout

Turn off writing to stdout. Useful for many iterations eg. in large simulations

3.2 dosurvey.py

-f <filename>

Input population model to use (default=populate.model)

-surveys <SURVEY NAME(S)>

Required: Name(s) of surveys to simulate on the population model

--noresults

By default, a .results file is written, containing a model of the population detected in the survey. This option switches off the writing of this file.

--asc

Write the survey model in plain ascii (psrpop old style). Not recommended, since the cPickle '.results' file is easier to work with.

--summary

Write a short .summary file (per survey) describing number of detections, number of pulsars outside survey area, number smeared, and number not beaming

--nostdout

Turn off writing to stdout. Useful for many iterations eg. in large simulations

3.3 view.py

-f <input model>

Select the population model to view (default=populate.model)

-p <param name>

Select the population parameter to plot

Supported: 'period', 'dm', 'gl', 'gb', 'lum', 'alpha', 'r0', 'rho', 'width', 'spindex', 'scindex', 'dist'

--logx

Plot log10 of the values

3.4 visualize.py

-f <model>

Select a population model to plot (default = populate.model)

-frac <F>

Plot a fraction <F> of the population for speed gains

CHAPTER

FOUR

TUTORIAL - BASIC USAGE

This page will outline a very simple pipeline for basic population simulations with PsrPopPy.

4.1 Generate Population Model

Population models are generated using the `populate` script (installed automatically with pip). A common use would be to generate a population of normal pulsars using the Parkes Multibeam Pulsar Survey as a basis. This survey detected 1038 pulsars (at last count). Using default radial distribution, period and luminosity models, we can generate such a population using the command:

```
populate -n 1038 -surveys PMSURV
```

This will then run for a few minutes, until the model PMSURV survey has detected 1038 pulsars. The file `populate.model` will be produced by default, which is a pickled population object.

If, instead, you wanted to use the Lyne & Manchester (1998) electron distribution, and for whatever reason wanted to store the output in the file `pop_lm98.model`, then we could run:

```
populate -n 1038 -surveys PMSURV -dm lm98 -o pop_lm98.model
```

A different output file will then be produced, where the population uses the new simulation parameters.

4.2 Simulate a Pulsar Survey

Once you've generated a pulsar population model, the `dosurvey` script can be used to run the model through a past, present or future, pulsar survey (as specified in files in the `survey` directory — see _model-survey-files).

For example, say we want to take the population model we just created, `pop_lm98.model`, and estimate from this how many pulsars would be detected in a putative LOFAR pulsar survey. This can be simply done using:

```
dosurvey -f pop_lm98.model -surveys LOFAR
```

4.3 Visualising a Pulsar Model

There are two ways to visualise the populations generated by either `populate.py` (`.model`) or `dosurvey.py` (`.results`). To plot basic histograms of various parameters, use the `view.py` script:

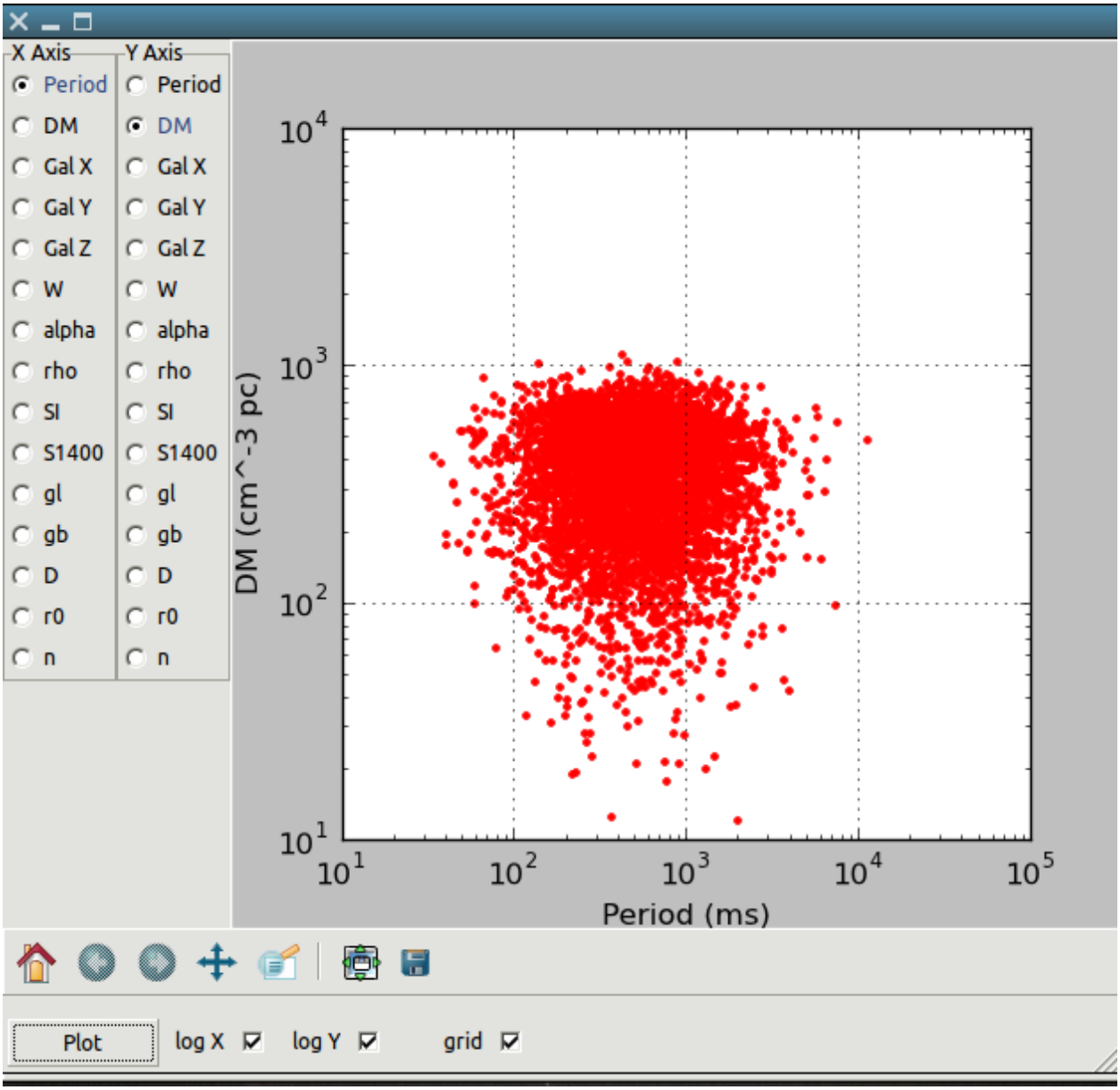
```
python view.py -f <model> -p <parameter>
```

Here <parameter> could be `period`, `dm`, or several other options, as outlined in _view_doc. Assuming the Matplotlib package is installed, this will generate a histogram which can then be saved or printed as necessary.

To create a histogram of the logarithm of the selected parameter, use:

```
python view.py -f <model> -p <parameter> --logx
```

If you have the wxPython plugin working (seems on newer macs this is a non-trivial piece of software to install — using macports is recommended) then use `wxView.py` to create scatter plots of various parameters (see screenshot below).



CHAPTER

FIVE

MODULE-LEVEL DOCUMENTATION

5.1 pulsar – Creates/stores a pulsar object

class pulsar.**Pulsar**

Pulsar.**__init__**([*period*, *dm*, *gl*, *gb*, *galCoords*, *r0*, *dtrue*, *lum_1400*, *spindex*, *alpha*, *rho*, *width_degree*, *snr*, *beaming*, *scindex*, *gpsFlag*, *gpsA*, *brokenFlag*, *brokenSI*])

Initialise the pulsar object

Pulsar.**s_1400**()

Returns the flux at 1400 MHz, calculated as

$S_{1400} = \frac{L_{1400}}{D^2_{\rm true}}$

Pulsar.**width_ms**()

Returns the pulse width in milliseconds, calculated as

$W_{\rm ms} = P_{\rm ms} \times W_{\rm degree}/360$

5.2 population – Creates/stores a population

class population.**Population**

Population.**__init__**([*pDistType*, *radialDistType*, *lumDistType*, *pmean*, *psigma*, *simean*, *sisigma*, *lummean*, *lumsigma*, *zscale*, *electronModel*, *gpsFrac*, *gpsA*, *brokenFrac*, *brokenSI*, *ref_freq*])

Initialise the population object

Population.**__str__**()

Defines how the operation `print Population` is performed

Population.**size**()

Returns the number of pulsars in the population object

Population.**join**(*poplist*)

Joins each of the populations in list `poplist` to the current population

Population.**write**(*outf*)

Uses cPickle to dump the population to file `outf`

Population.**write_asc**(*outf*)

Writes the population to an ascii file in the old psrpop way

5.3 survey – Read a survey file into a survey object

class survey.**Survey**

Survey.**__init__**(*surveyName*)

Read in a (correctly formatted!) survey file

Survey.**__str__**()

Define how to perform `print Survey`

Survey.**nchans**()

Returns the number of channels, calculated as

$n_{\rm chans} = \frac{\rm BW_{total}}{\rm BM_{chan}}$

Survey.**inRegion**(*pulsar*)

Determines if `Pulsar` is inside survey region. Returns True or False accordingly

Survey.**inPointing**(*pulsar*)

Determines if `Pulsar` is inside one of the survey's `pointings`. Returns the offset from beam centre to the pulsar.

Survey.**SNRcalc**(*pulsar*, *pop*)

Calculates the SNR of a `Pulsar` from `Population` `pop` in the survey. Returns -1 if pulse is smeared, and -2 if pulsar is outside survey region. SNR is calculated (with familiar terms) as

$$\rm SNR = \frac{S_{1400} G \sqrt{n_{pol} BW \tau}}{\beta T_{tot}} \sqrt{\frac{1-\delta}{\delta}} \eta$$

where

$$\eta = \exp(-2.7727 \times \rm offset^2 / fwhm^2)$$

class survey.**Pointing**

Pointing.**__init__**(*coord1*, *coord2*, *coordtype*)

Converts (coord1, coord2) into correctly formatted (l, b). Coordtype must be either `eq` or `gal`. If `eq`, the RA and Dec are converted internally

5.4 populate – Create a population object

class populate.**Populate**

Populate.**generate**(*ngen*[, *surveyList*, *pDistType*, *radialDistType*, *electronModel*, *pDistPars*, *siDistPars*, *lumDistType*, *lumDistPars*, *zscale*, *duty*, *scindex*, *gpsArgs*, *doubleSpec*, *nostdout*])

The method called by the `populate.py` command-line-script

Populate.**write**(*outf=populate.model*)

Writes the `Population` model into file outf as a cPickle dump

5.5 radialmodels – Container for radial distn models

class radialmodels.**RadialModels**

radialmodels.**seed**()

Call the FORTRAN routine to make a seed

radialmodels.**slabdist**()
Pick a point from a "slab" distribution around the Galactic plane

radialmodels.**diskdist**()
Pick a point from a distribution purely along the Galactic plane

radialmodels.**lfl06**()
Pick a point from the Lorimer et al (2006) Galactic distribution

radialmodels.**ykr**()
Pick a point from the Yusifov & Kucuk Galactic distribution

5.6 *galacticops* – Container for functions relating to the Galaxy

class galacticops.**GalacticOps**

galacticops.**calc_dtrue**(*(x, y, z)*)
Calculate the distance from the Sun to Galactic coords (x, y, z) (NB. tuple)

galacticops.**calcXY**(*r0*)
Given a Galactic radius r0, choose an (x, y) position at random θ

galacticops.**ne2025_dist_to_dm**(*dist*, *gl*, *gb*)
Given a distance and Galactic coordinates, calculate DM according to NE2025

galacticops.**ne2001_dist_to_dm**(*dist*, *gl*, *gb*)
Given a distance and Galactic coordinates, calculate DM according to NE2001

galacticops.**lmt85_dist_to_dm**(*dist*, *gl*, *gb*)
Given a distance and Galactic coordinates, calculate DM according to the LMT85 model

galacticops.**lb_to_radec**(*gl*, *gb*)
Convert Galactic coordinates to equatorial

galacticops.**ra_dec_to_lb**(*ra*, *dec*)
Convert equatorial coordinates to Galactic

galacticops.**tsky**(*gl*, *gb*, *freq*)
Calculate sky temperature at observing frequency freq and at Galactic coordinates gl, gb according to Haslam et al

galacticops.**xyz_to_lb**(*(x, y, z)*)
Convert the tuple (x, y, z) to Galactic sky coordinates.

Returns l, b in degrees

galacticops.**lb_to_xyz**(*l*, *b*, *dist*)
Convert Galactic sky coordinates at a distance dist to x,y,z coordinates.

Returns position as a tuple

galacticops.**scatter_bhat**(*dm*, *scatterindex*, *freq_mhz*)
Calculate the scatter time according to Bhat et al at. Frequency in MHz, pulsar with dispersion measure dm, and using a scattering spectral index of scatterindex.

Calculated as

$\tau = -6.46 + 0.154 \log_{10}(\mathrm{dm}) + 1.07 \log_{10}(\mathrm{dm})^2 + \mathrm{scatterindex} \times \log_{10}(\frac{\mathrm{freq\_mhz}}{1000})$

and typically $\mathrm{scatterindex} = -3.86$ (but there is an option to vary it!)

CHAPTER

SIX

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PYTHON MODULE INDEX

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